Aeroelastic Vibration Analysis of a Finite-Span NACA 2412 Wing

A Strip-Theory Coupled Structural–Aerodynamic Formulation

1 Problem Setup

The wing is idealized as a straight cantilever beam clamped at the root ($y = 0$) and free at the tip ($y = L$), where L is the semi-span. Each spanwise station has two elastic degrees of freedom: a plunge (bending) deflection $h(y, t)$, positive downward, measured at the elastic axis, and a pitch (twist) rotation $\theta(y, t)$, positive nose-up, about the same elastic axis. The chord distribution is $c(y)$ and the section is a NACA 2412 profile at every station (no spanwise twist or taper assumed, though the formulation supports both). The wing flies at freestream speed U and air density ρ, with dynamic pressure $q_\infty = \frac{1}{2}\rho U^2$.

The objective is to determine how the wing's vertical (plunge-dominated) vibration behaves as U is varied, for a wing of finite span L (finite aspect ratio $AR = b^2/S$, where $b = 2L$ is the full span and S the planform area).

2 Structural Model

2.1 Continuous governing equations

Euler–Bernoulli bending and Saint-Venant torsion of a non-uniform beam, each driven by the distributed aerodynamic load:

$$\frac{\partial^2}{\partial y^2}\left(EI(y)\,\frac{\partial^2 h}{\partial y^2}\right) + m(y)\,\frac{\partial^2 h}{\partial t^2} = L'(y,t), \tag{1}$$

$$-\frac{\partial}{\partial y}\left(GJ(y)\,\frac{\partial \theta}{\partial y}\right) + I_\theta(y)\,\frac{\partial^2 \theta}{\partial t^2} = M'_{ea}(y,t), \tag{2}$$

where $EI(y)$ is bending stiffness, $GJ(y)$ torsional stiffness, $m(y)$ mass per unit length, $I_\theta(y)$ mass moment of inertia per unit length about the elastic axis, $L'(y, t)$ the aerodynamic lift per unit span, and $M'_{ea}(y, t)$ the aerodynamic pitching moment per unit span about the elastic axis.

2.2 Finite element discretization

The span is divided into N_e two-node beam elements of length L_e. Each node carries three degrees of freedom: $\{w,\ w',\ \theta\}$ (transverse deflection, slope, twist). For a uniform element the classical consistent mass/stiffness sub-matrices are

$$\boldsymbol{K}^{(e)}_{\text{bend}} = \frac{EI}{L_e^3}\begin{bmatrix} 12 & 6L_e & -12 & 6L_e \\ 6L_e & 4L_e^2 & -6L_e & 2L_e^2 \\ -12 & -6L_e & 12 & -6L_e \\ 6L_e & 2L_e^2 & -6L_e & 4L_e^2 \end{bmatrix}, \qquad \boldsymbol{M}^{(e)}_{\text{bend}} = \frac{mL_e}{420}\begin{bmatrix} 156 & 22L_e & 54 & -13L_e \\ 22L_e & 4L_e^2 & 13L_e & -3L_e^2 \\ 54 & 13L_e & 156 & -22L_e \\ -13L_e & -3L_e^2 & -22L_e & 4L_e^2 \end{bmatrix} \tag{3}$$

acting on $\{w_1, w_1', w_2, w_2'\}$, and the rod-type torsion sub-matrices

$$\boldsymbol{K}^{(e)}_{\text{tors}} = \frac{GJ}{L_e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \boldsymbol{M}^{(e)}_{\text{tors}} = \frac{I_\theta L_e}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \tag{4}$$

acting on $\{\theta_1, \theta_2\}$. Bending and torsion are structurally uncoupled (elastic axis = reference axis), so the two blocks are merged into a single 6×6 element matrix ordered $\{w_1, w_1', \theta_1, w_2, w_2', \theta_2\}$ and assembled into global matrices $\boldsymbol{M}_s, \boldsymbol{K}_s$ by standard FEM assembly. The three root DOFs are removed (cantilever condition $w = w' = \theta = 0$ at $y = 0$), leaving a reduced DOF vector $\boldsymbol{q}(t)$ of length $3N_e$ (minus 3 removed root DOFs +3 trailing-edge-of-root node accounting – i.e. $3N_e$ free DOFs after elimination).

3 Aerodynamic Model: Quasi-Steady Strip Theory

3.1 Local effective incidence

At spanwise station y, the local plunge velocity $\dot{h}(y,t)$ adds an induced angle of attack, so the effective incidence seen by the strip is

$$\alpha_{\text{eff}}(y,t) = \theta(y,t) + \frac{\dot{h}(y,t)}{U}. \tag{5}$$

This is the baseline model implemented below and in the accompanying code. A complete quasi-steady treatment also includes a pitch-*rate*-dependent circulatory lift term (Theodorsen's quasi-steady limit adds $\left(\frac{1}{2} - a\right)\frac{c(y)}{U}\dot{\theta}(y,t)$ inside α_{eff}, where a locates the elastic axis as a fraction of chord from mid-chord). That term's contribution to the lift is physically distinct from the angle-of-attack term above, and critically, it does *not* act through the same moment arm $e(y)$ used in Eq. (7) below – its own moment arm about the elastic axis requires separate, careful derivation (see Theodorsen 1935, or Hodges & Pierce, *Introduction to Structural Dynamics and Aeroelasticity*). Rather than include that term on unverified footing, it is deliberately omitted from the baseline formulation and code; Section 7 notes the resulting limitation and a safe way to work around it.

3.2 Section lift and moment

$$L'(y,t) = q_\infty\, c(y)\, C_{L\alpha}\, \alpha_{\text{eff}}(y,t), \tag{6}$$

$$M'_{ea}(y,t) = e(y)\, L'(y,t) \;+\; q_\infty\, c(y)^2\, C_{m,ac}, \tag{7}$$

where $e(y)$ is the offset of the aerodynamic center (quarter-chord) ahead of the elastic axis, $C_{L\alpha}$ the lift-curve slope, and $C_{m,ac}$ the (camber-induced, load-independent) moment coefficient about the aerodynamic center. The $C_{m,ac}$ term is a constant, non-linear-in-$(q, \dot{q})$ static twisting moment; it contributes to the mean trim twist but *not* to the linear dynamic stability matrices below, and is tracked separately.

3.3 Finite-span correction

Equation (6) uses the 2D (infinite-span) lift-curve slope. For a finite wing of aspect ratio AR and span efficiency e_s (Oswald efficiency factor), the effective 3D lift-curve slope from Prandtl lifting-line theory is

$$C_{L\alpha}^{3D} = \frac{C_{L\alpha}^{2D}}{1 + \dfrac{C_{L\alpha}^{2D}}{\pi\, AR\, e_s}}. \tag{8}$$

This reduced slope is used uniformly in Eq. (6) as a first-order finite-span correction; a spanwise-varying correction (e.g. via a vortex-lattice induced-angle solve) is a natural refinement.

4 Aeroelastic Coupling: Generalized Equations of Motion

The FEM shape functions interpolate $h(y,t)$ and $\theta(y,t)$ from the nodal DOF vector $\boldsymbol{q}(t)$: $h(y,t) = \boldsymbol{N}_w(y)\,\boldsymbol{q}(t)$, $\theta(y,t) = \boldsymbol{N}_\theta(y)\,\boldsymbol{q}(t)$.

Sign convention. h (and the nodal DOF w) is positive *downward* (Section 1), while L' is the physical lift, positive *upward* as is standard in aerodynamics. The generalized force conjugate to w is therefore $-L'$: an upward lift resists an increase in downward deflection. This is the classical typical-section convention (e.g. Hodges & Pierce; Wright & Cooper), $m\ddot{h} + Kh = -L'$, together with $I_\theta\ddot{\theta} + K\theta = +M'_{ea}$ for the moment equation (θ positive nose-up, no sign flip needed there).

The spanwise-distributed aerodynamic loads of Eqs. (6)–(7) are linear in $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$, so the generalized aerodynamic force vector, lumped node-by-node (one aerodynamic "strip" per node, matching the discretization request this document responds to) rather than integrated against the full FEM shape functions, can be written as

$$\boldsymbol{Q}_{\text{aero}}(t) = -\,\boldsymbol{D}_a(U)\,\dot{\boldsymbol{q}}(t) \;-\; \boldsymbol{S}_a(U)\,\boldsymbol{q}(t) \;+\; \boldsymbol{Q}_{\text{static}}, \tag{9}$$

with, at node i (local block in the $\{w_i, w'_i, \theta_i\}$ ordering, all other entries zero),

$$\boldsymbol{S}_a^{(i)} = \begin{bmatrix} 0 & 0 & +L'_{,\theta} \\ 0 & 0 & 0 \\ 0 & 0 & -M'_{,\theta} \end{bmatrix} \Delta y_i, \qquad \boldsymbol{D}_a^{(i)} = \begin{bmatrix} +L'_{,\dot{h}} & 0 & 0 \\ 0 & 0 & 0 \\ -M'_{,\dot{h}} & 0 & 0 \end{bmatrix} \Delta y_i, \tag{10}$$

where $L'_{,\theta} = q_\infty c\, C_{L\alpha}^{3D}$, $L'_{,\dot{h}} = q_\infty c\, C_{L\alpha}^{3D}/U$, and $M'_{,(\cdot)} = e(y)\, L'_{,(\cdot)}$ (from Eq. (7), dynamic part only), and Δy_i is node i's tributary span width (half of each adjacent element). $\boldsymbol{D}_a$ is the *aerodynamic damping matrix* and $\boldsymbol{S}_a$ the *aerodynamic stiffness matrix*; both scale with $q_\infty \propto U^2$ except for the $\dot{h}/U$-based entries, which scale linearly in U. The slope DOF w' carries no direct aerodynamic generalized force in this lumped, strip-per-node treatment – a consistent shape-function-weighted integral would add one; this is a deliberate simplification matching the "divide the span into sections" formulation requested.

The coupled linear equations of motion are then

$$\boldsymbol{M}_s\,\ddot{\boldsymbol{q}} + \big(\boldsymbol{C}_s + \boldsymbol{D}_a(U)\big)\,\dot{\boldsymbol{q}} + \big(\boldsymbol{K}_s + \boldsymbol{S}_a(U)\big)\,\boldsymbol{q} = \boldsymbol{0}, \tag{11}$$

where $\boldsymbol{C}_s$ is structural damping (often a small Rayleigh damping matrix, $\boldsymbol{C}_s = \alpha\boldsymbol{M}_s + \beta\boldsymbol{K}_s$).

5 Flutter / Stability Analysis (p-method)

Assume a solution of the form $\boldsymbol{q}(t) = \hat{\boldsymbol{q}}\, e^{st}$. Substitution into Eq. (11) gives the quadratic eigenvalue problem

$$\Big[s^2\boldsymbol{M}_s + s\big(\boldsymbol{C}_s + \boldsymbol{D}_a(U)\big) + \big(\boldsymbol{K}_s + \boldsymbol{S}_a(U)\big)\Big]\hat{\boldsymbol{q}} = \boldsymbol{0}, \tag{12}$$

solved at each trial airspeed U (e.g. via linearization to a first-order state-space form and a standard eigenvalue solver). Each eigenvalue $s_k(U) = \sigma_k(U) \pm i\,\omega_k(U)$ gives, at that speed, a modal frequency $f_k = \omega_k/2\pi$ and a damping ratio $g_k = -\sigma_k/\omega_k$ (or equivalently σ_k itself).

- **V–f diagram**: $\omega_k(U)/2\pi$ vs. U for each mode; flutter is indicated where two modal frequencies approach one another (coalescence).
- **V–g diagram**: $\sigma_k(U)$ (or g_k) vs. U; the *flutter speed* U_F is the lowest U at which any σ_k crosses zero from negative (damped) to positive (unstable).
- **Torsional divergence** is a static ($\omega = 0$) instability and shows up separately as $\det(\boldsymbol{K}_s + \boldsymbol{S}_a(U)) \to 0$.

In the implementation, the complete state-space spectrum is retained by default rather than truncating the search to a prescribed number of modes. Oscillatory branches are continued between adjacent airspeeds using eigenvector modal assurance criterion (MAC) correlation. If a branch becomes real or overdamped, its oscillatory frequency and growth rate are recorded as unavailable (NaN) rather than being replaced by another mode. The real eigenvalues are retained and plotted separately. Dynamic instability is reported directly from

$$\sigma_{\max}(U) = \max_k \Re(s_k(U)), \tag{13}$$

using the complete spectrum, including real eigenvalues; the first speed where $\sigma_{\max} \geq 0$ is the reported instability speed. Static divergence is checked independently using the normalized indicator

$$d_D(U) = \frac{\det(\boldsymbol{K}_{s,r} + \boldsymbol{S}_{a,r}(U))}{\det(\boldsymbol{K}_{s,r})}, \tag{14}$$

where the subscript r denotes the free (cantilever) DOFs. A sign change in d_D brackets a divergence speed; the determinant indicator is not used as a substitute for the dynamic flutter test.

For $U < U_F$, forced response to gusts/turbulence (replacing the RHS of Eq. (11) with a stochastic or deterministic gust-force vector) gives the finite vertical vibration amplitude at the tip, root, or any station of interest; this amplitude grows sharply as $U \to U_F$ because the effective damping of the governing mode approaches zero.

The companion code also provides a prescribed vertical-gust response. For a gust velocity $w_g(t)$, the additional local incidence is $\alpha_g = w_g/U$, giving

$$L'_g(y,t) = q_\infty c(y) C^{3D}_{L\alpha}\frac{w_g(t)}{U}, \qquad Q_{w,g} = -L'_g\Delta y, \qquad Q_{\theta,g} = e(y)L'_g\Delta y. \tag{15}$$

These generalized forces are assembled node-by-node and applied on the right-hand side of the time-domain equations. The supplied example uses a sinusoidal gust; the same interface accepts an impulse, one-minus-cosine gust, or stochastic gust function.

6 NACA 2412 Aerodynamic Data

- Thin-airfoil lift-curve slope: $C_{L\alpha}^{2D} = 2\pi \text{ rad}^{-1}$ (theoretical); experimentally, finite-thickness/viscous effects typically give $C_{L\alpha}^{2D} \approx 5.7$–$6.3 \text{ rad}^{-1}$ at low Mach, pre-stall.
- Zero-lift angle of attack (due to camber): $\alpha_0 \approx -2.1^\circ$.
- Moment coefficient about the aerodynamic center (quarter-chord), essentially constant pre-stall: $C_{m,ac} \approx -0.045$ to -0.05.
- Aerodynamic center location: ≈ 0.24–$0.25\,c$ from the leading edge.

These feed directly into Eqs. (6)–(7) as the per-section aerodynamic properties, uniform along the span if the wing is untapered and untwisted.

7 Numerical Implementation Correspondence

The companion MATLAB code mirrors this document one section at a time:

1. `naca2412_data.m` — Section 6 aerodynamic coefficients.
2. `beam_fem_matrices.m` — Section 2.2, assembling $\boldsymbol{M}_s, \boldsymbol{K}_s$.
3. `aero_strip_matrices.m` — Section 3–4, building $\boldsymbol{D}_a(U)$, $\boldsymbol{S}_a(U)$ from strip integration with the finite-span correction of Eq. (8).
4. `flutter_sweep.m` — Section 5, sweeping U, solving Eq. (12), retaining the complete spectrum, MAC-tracking oscillatory modes, separating real eigenvalues, checking the static divergence indicator, and returning the dominant eigenvalue.
5. `gust_response.m` — prescribed vertical-gust forcing and time-domain integration of the coupled system below a selected flight speed.

The parameter a_{ea} is passed through the aerodynamic interface but is unused by the baseline model. It would enter the omitted pitch-rate circulatory contribution $(1/2 - a_{\text{ea}})c\dot{\theta}/U$. That contribution has a distinct aerodynamic moment arm and is therefore not inserted using the angle-of-attack moment arm $e(y)$ without a separate checked derivation.

8 MATLAB Code Block Guide

This section describes the role of the principal code blocks in the MATLAB files and their correspondence with the equations above.

8.1 `naca2412_data.m`: section data

The function `naca2412_data` creates the `aero` structure used by the aerodynamic routines. The fields are:

- `Cl_alpha_2D`: the two-dimensional lift-curve slope;
- `alpha0`: the measured zero-lift angle of attack;

- `Cm_ac`: the quarter-chord pitching-moment coefficient;
- `ac_loc`: the aerodynamic-center chordwise location.

The current linear stability matrices use `Cl_alpha_2D`, `Cm_ac`, and `ac_loc`. The `alpha0` value is stored for static trim and extended section-load calculations but is not included in the present dynamic matrices.

8.2 `beam_fem_matrices.m`: structural assembly

The input block defines the span mesh and evaluates any spanwise-varying structural properties at each element midpoint. For every two-node element, the code then:

1. constructs the 4×4 Euler–Bernoulli bending stiffness and consistent mass matrices;
2. constructs the 2×2 rod-type torsional stiffness and mass matrices;
3. embeds these submatrices in a 6×6 element matrix ordered as $\{w_1, w_1', \theta_1, w_2, w_2', \theta_2\}$;
4. scatters the element matrices into the global $\boldsymbol{K}_s$ and $\boldsymbol{M}_s$ matrices.

The final block removes the three root DOFs and returns `free_dofs`, which is used to form the cantilever system for all subsequent calculations.

8.3 `aero_strip_matrices.m`: strip loads

This function first evaluates

$$C_{L\alpha}^{3D} = \frac{C_{L\alpha}^{2D}}{1 + C_{L\alpha}^{2D}/(\pi A R e_s)} \tag{16}$$

and computes a tributary span width for every node. The loop over nodes evaluates the local chord and the aerodynamic-center-to-elastic-axis offset $e = (x_{ea} - x_{ac})c$. It then forms the lift sensitivities

$$\frac{\partial L}{\partial \theta} = q_\infty c C_{L\alpha}^{3D} \Delta y, \qquad \frac{\partial L}{\partial \dot{w}} = \frac{q_\infty c C_{L\alpha}^{3D}}{U} \Delta y. \tag{17}$$

These sensitivities are placed into the local 3×3 aerodynamic stiffness and damping blocks. The slope DOF w' receives no direct load in this lumped strip approximation. The function also returns `Qstatic`, which contains the constant camber-moment contribution and is separate from the linear stability matrices.

8.4 `flutter_sweep.m`: eigenvalue sweep and tracking

For each airspeed, the function:

1. reduces $\boldsymbol{M}_s$, $\boldsymbol{K}_s$, and $\boldsymbol{C}_s$ to the free DOFs;
2. obtains $\boldsymbol{D}_a(U)$ and $\boldsymbol{S}_a(U)$ from `aero_strip_matrices`;
3. forms the state matrix

$$\boldsymbol{A}(U) = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{M}_r^{-1}(\boldsymbol{K}_r + \boldsymbol{S}_{a,r}) & -\boldsymbol{M}_r^{-1}(\boldsymbol{C}_r + \boldsymbol{D}_{a,r}) \end{bmatrix}; \tag{18}$$

4. computes the complete eigenvalue set $s_k(U)$;

5. stores the eigenvalue with the largest real part as the dominant stability eigenvalue;
6. separates real/overdamped eigenvalues from oscillatory eigenvalues;
7. uses eigenvector MAC correlation to continue oscillatory branches between adjacent airspeeds, leaving an unmatched branch as `NaN`;
8. evaluates the normalized static-divergence determinant indicator.

The returned `max_growth` is $\max_k \Re(s_k)$ over the complete spectrum and is therefore the quantity used for the first-instability check.

8.5 `run_flutter_analysis.m`: top-level workflow

The script is organized into five stages:

1. define geometry, structural properties, damping, and the airspeed grid;
2. construct the NACA data and structural matrices;
3. run the aeroelastic eigenvalue sweep;
4. report the first complete-spectrum instability and the separate divergence diagnostic;
5. plot the V–f, V–g, real-eigenvalue, and static-divergence diagrams.

The final blocks select a check speed, integrate an initial-condition response using `ode45`, and then call `gust_response` for a prescribed sinusoidal vertical gust.

8.6 `gust_response.m`: forced response

The gust function converts a prescribed vertical gust velocity $w_g(t)$ into an additional incidence $\alpha_g = w_g/U$. At every node it forms the gust lift $L'_g = q_\infty c C^{3D}_{L\alpha} \alpha_g \Delta y$ and assembles the generalized force components

$$Q_{w,g} = -L'_g, \qquad Q_{\theta,g} = eL'_g. \tag{19}$$

The reduced forced state equation is then integrated with `ode45`. The returned `F_hist` records the applied generalized gust force at the solver output times.

Note on validation. This is a first-principles quasi-steady strip-theory template, not a pre-validated production tool. The matrix assembly was checked with an independent re-implementation (structural $\boldsymbol{M}_s, \boldsymbol{K}_s$ symmetric and positive definite on the free DOFs; the lowest cantilever bending/torsion frequencies match beam-theory expectations; the plunge-dominated mode is stably damped at low airspeed and becomes unstable only above a finite crossing speed, matching the expected single-mode-becomes-unstable flutter pattern). Known limitations to be aware of before trusting absolute numbers:

- **No torsional aerodynamic damping term** is included (Section 3.1): the torsion-dominated mode can show a very weak instability even at low airspeed if structural damping $\boldsymbol{C}_s$ is left at exactly zero. `run_flutter_analysis.m` defaults to a small nonzero Rayleigh β specifically to avoid this; replace it with measured/estimated structural damping, or implement the omitted pitch-rate circulatory term from a checked reference, for quantitative work.
- No unsteady lag (Theodorsen's $C(k)$) or finite-span-varying induced angle (vortex-lattice) is included – both are noted as natural refinements in Sections 3.3 and 7's companion code.

- The gust response is a prescribed external-forcing calculation; it does not include gust alleviation, unsteady aerodynamic lag, or a turbulence spectrum unless those are supplied through the gust-velocity function.

Sanity-checking further against a classical closed-form case (e.g. Theodorsen's flat-plate typical-section result) or a published NACA 2412 wind-tunnel/flutter test is still recommended before using this for anything beyond a qualitative/teaching tool.